

Medical Insurance Charge Predictor

Problem Statement

The objective is to build a **regression model** that predicts **medical insurance charges** based on customer attributes such as:

- Age
- Sex
- BMI
- Number of children
- Smoking status

This is a, Machine Learning -> Supervised Learning -> Regression

Basic Information About Dataset

Age	Age of primary beneficiary
Sex	Gender
BMI	Body Mass Index
Children	Number of dependent children
Smoker	Smoking status
Charges	Target : Medical Insurance cost

Data Preprocessing

Converting string to numeric	Sex : Male = 0, Female = 1 Smoker : Yes = 1, No = 0
Scaling	Standard Scalar
Train-Test Split	70% training, 30% testing

Multiple Model Analysis

1. Multiple Linear Analysis

0.7894790349867009

2. Support Vector Machine

HYPER PARAMETER	LINEAR	RBF	POLY	SIGMOID
C = 10	0.46246841423396834	-0.0322732939067103	0.038716222760231456	0.03930714378274347
C = 100	0.6288792857320361	0.3200317832050832	0.6179569624059796	0.5276103546510407
C = 1000	0.7649311738597486	0.8102064874808204	0.8566487675946555	0.2874706948697727
C = 2000	0.7440418308107889	0.8547766422240716	0.8605579276143769	-0.5939509731283508
C = 3000	0.7414236599248987	0.8663393963090398	0.8598930084494327	r score is deteriorating, no need to test
C = 5000	0.7414179301703103	0.874777817423661	0.8595656374126361	r score is deteriorating, no need to test

3. Decision Tree

CRITERION	SPLITTER	R VALUE
squared_error	best	0.6935643620288746
squared_error	random	0.6786146385639003
friedman_mse	best	0.6851078940595519
friedman_mse	random	0.6677158531404498
absolute_error	best	0.6602322905539526
absolute_error	random	0.7455071283652015
poisson	best	0.7201600296042935
poisson	random	0.7013998594969812

4. Decision Tree

CRITERION	SPLITTER	R VALUE
squared_error	n_estimators = 50	0.8495860472309916
squared_error	n_estimators = 100	0.8537074492312178
absolute_error	n_estimators = 50	0.8533104199010396
absolute_error	n_estimators = 100	0.8522171666048011
friedman_mse	n_estimators = 50	0.8498058213339406
friedman_mse	n_estimators = 100	0.8540807721486975
poisson	n_estimators = 50	0.8491113222296434
poisson	n_estimators = 100	0.8526481325996583

5. Ada Boosting (Regression)

0.8618083410771146

6. Gradient Boosting (Regression)

0.8833974199848822

Final Outcome

Gradient Boosting (Regression) algorithm is giving the best results.
Still, couldn't achieve 90+, this is the best we've got